

CORONARY, PERIPHERAL, AND STRUCTURAL INTERVENTIONS

THE FOUR CORNERS: CLINICAL VIGNETTE CORNER

Coronary Pseudoaneurysm and Stent Disruption Caused by Infective Endocarditis



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ABSTRACT

IMAGES Intravascular imaging (optical coherence tomography and intravascular ultrasound) demonstrating coronary pseudoaneurysm and stent disruption.

CASE SUMMARY A 65-year-old man undergoing hemodialysis for diabetic nephropathy developed methicillin-resistant *Staphylococcus aureus* infective endocarditis complicated by coronary pseudoaneurysm and stent disruption, resulting in cardiac tamponade. Intravascular imaging confirmed the complete destruction of the arterial wall. Emergency covered stent implantation and pericardiocentesis stabilized the patient, followed by a multidisciplinary treatment approach, including prolonged antibiotics and surgical source control, leading to survival.

TAKE-HOME MESSAGE Timely imaging-guided diagnosis and a stepwise multidisciplinary strategy were lifesaving in this extremely rare and complex case. (JACC Case Rep. 2026;31:106337) © 2026 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

A 65-year-old man on hemodialysis for diabetic nephropathy, with a history of leadless pacemaker implantation and prior percutaneous coronary intervention with drug-eluting stenting of the left anterior descending artery for stable angina (Figure 1A), developed methicillin-resistant *Staphylococcus aureus* bacteremia complicated by infective endocarditis.

During intravenous antibiotic therapy for methicillin-resistant *Staphylococcus aureus* bacteremia, he developed chest pain. Echocardiography revealed new left ventricular wall motion abnormalities and

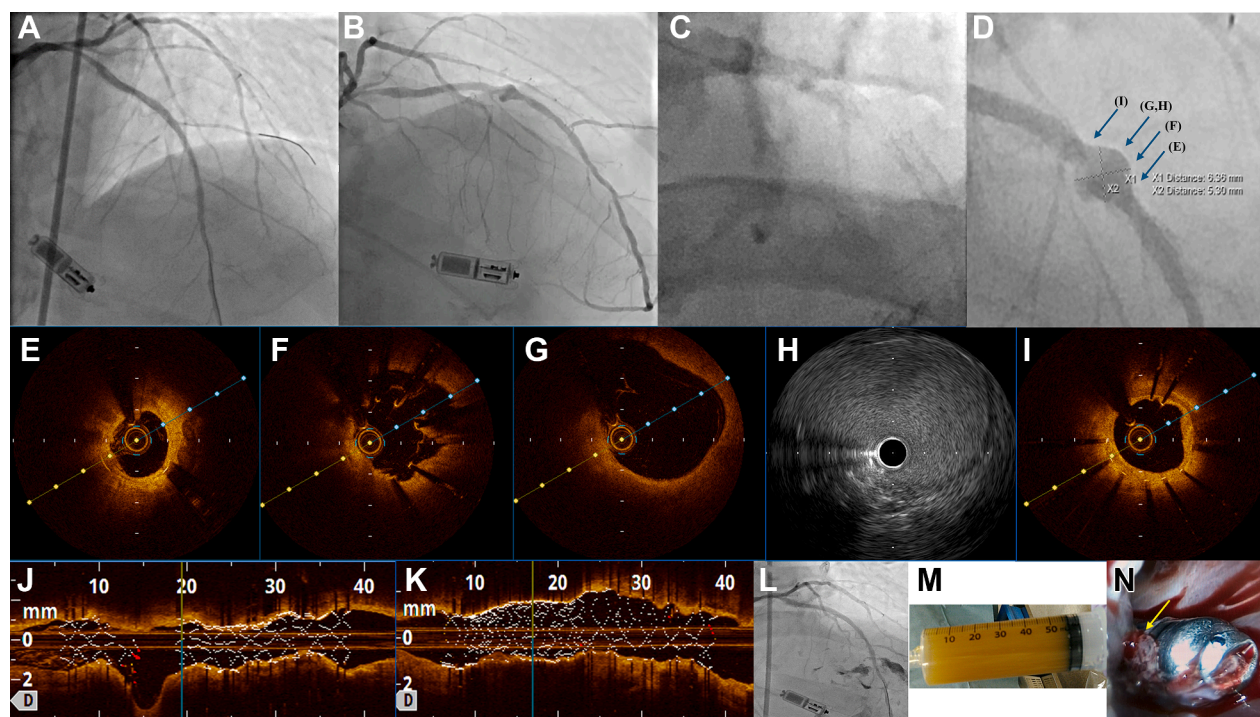
TAKE-HOME MESSAGES

- This report describes an extremely rare and complex presentation of a coronary pseudoaneurysm and stent disruption complicated by infective endocarditis in a patient with multiple comorbidities (diabetes, end-stage renal disease, and prior device implantation).
- Survival was achieved through prompt intravascular imaging-guided diagnosis and a multidisciplinary, stepwise treatment strategy including pericardiocentesis, covered stent placement, prolonged antibiotics, and surgical source control.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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FIGURE 1 Multimodality Imaging and Covered Stent Management of a Coronary Pseudoaneurysm With Infective Endocarditis

(A) Baseline angiography showing prior drug-eluting stent in the left anterior descending artery. (B to D) Coronary angiography after pericardiocentesis demonstrating stent disruption with a rapidly enlarging pseudoaneurysm. (E to J) Optical coherence tomography showing loss of the three-layered structure, consistent with a pseudoaneurysm. (H) Intravascular ultrasound confirming pseudoaneurysm formation. (K) Covered stent deployed to seal the lesion. (L) Final angiography confirming complete exclusion of the pseudoaneurysm. (M) Yellowish, turbid pericardial fluid obtained during emergency pericardiocentesis. (N) Surgical exploration of the leadless pacemaker revealing vegetation.

pericardial effusion. The patient rapidly developed hemodynamic instability owing to cardiac tamponade and cardiogenic shock. Laboratory tests revealed a high inflammatory response with disseminated intravascular coagulation, suggesting multiorgan dysfunction.

Following emergency pericardiocentesis, coronary angiography revealed disruption of the previously implanted stent and the presence of a rapidly enlarging coronary pseudoaneurysm (**Figures 1B to 1D**).

Optical coherence tomography (**Figures 1E to 1J**) and intravascular ultrasound (**Figure 1H**) revealed complete destruction of the arterial wall and loss of the normal 3-layer architecture, consistent with pseudoaneurysm. The pseudoaneurysm likely resulted from infectious degradation of the vessel wall, facilitated by septic microembolism or direct extension from bacteremia. Given the imminent risk of rupture, a covered coronary stent was deployed to seal the lesion (**Figure 1K**). Final angiography (**Figure 1L**) confirmed complete exclusion of the pseudoaneurysm. Yellowish, turbid pericardial fluid was aspirated immediately before percutaneous coronary intervention, further indicating a systemic infection (**Figure 1M**). After drainage and covered stent implantation, his vital signs and laboratory marker levels improved. The patient subsequently underwent coronary artery bypass grafting, leadless pacemaker removal, and epicardial pacemaker implantation. Surgical exploration of the leadless pacemaker revealed visible vegetation, suggesting device involvement (**Figure 1N**). He received a total of 4 months of hospitalization, including approximately 2 months of antibiotic therapy, and was eventually discharged home in stable condition. Coronary pseudoaneurysms caused by infective endocarditis are rare and life-threatening and lack standardized management protocols.¹ Surgical management of active infective endocarditis with impending pseudoaneurysm rupture can be extremely challenging and is often associated with poor outcomes, particularly when infection control is inadequate.² Therefore, timely diagnosis enabled by intravascular imaging and subsequent urgent treatment with covered stent

implantation proved critical for patient survival in this setting of impending rupture. This case highlights that even in the most complex clinical settings, rapid diagnosis and a stepwise multidisciplinary approach can be lifesaving.

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REFERENCES

1. Ramakumar V, Thakur A, Abdulkader RS, et al. Coronary stent infections - A systematic review and meta-analysis. *Cardiovasc Revasc Med*. 2023;54:16-24.
2. Pisani A, Braham W, Borghese O. Coronary stent infection: are patients amenable to surgical treatment? A systematic review and narrative synthesis. *Int J Cardiol*. 2021;344:40-46.

KEY WORDS endocarditis, imaging, percutaneous coronary intervention